SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Machine Learning

ASSESSMENT 3 – Technical Presentation (5 QUESTIONS)

Course Code:	ITA05	**Course Title:**	**Computer Vision**
CO Assessed:	CO1	**Assessment:**	Assessment 3: Technical Presentation
Weightage:	35% of CIA	**Total Marks:**	(5 Questions × 10 Marks)
CO1 Statement:	***Apply the fundamental concepts, image formation models, and image processing techniques to analyze and solve real-world Computer Vision problems.***		
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Section / Batch:	2024	**Date:**	14.10.2026

Code of Conduct

I VIMAL R (192421350) (NAME/ REG NO) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

QUESTIONS: 5 **Total Marks: 50**

Annexure A

Technical

Presentation

Q1. A shopping mall uses CCTV cameras that produce clear images during the day but dark and noisy images at night, making it difficult to identify people and objects.
Prepare a technical presentation demonstrating how filtering, noise removal, contrast enhancement, and sharpening techniques can be applied to improve the captured images. Present the sequence of image-processing operations required before object detection.

Q2. A university is developing an automated document-scanning system using a camera, where captured documents may contain shadows, noise, uneven illumination, tilted pages, and blurred text.
Prepare a technical presentation explaining how filtering, thresholding, contrast enhancement, edge detection, and segmentation can be applied to improve the quality of the document images. Demonstrate the role of each technique in the processing pipeline.

Q3. A fruit-processing company wants to automatically identify damaged apples from images captured on a conveyor belt, where shadows, reflections, different backgrounds, and brightness variations are present.
Prepare a technical presentation comparing global thresholding, adaptive thresholding, edge-based segmentation, and region-based segmentation for this application. Demonstrate how a suitable technique or combination of techniques can improve fruit defect detection.

Q4. An autonomous vehicle uses a front-facing camera to detect lane markings under different weather and lighting conditions, where shadows, damaged markings, vehicles, and rain may affect the captured images.
Prepare a technical presentation demonstrating how grayscale conversion, filtering, contrast enhancement, edge detection, thresholding, and region-of-interest techniques can be applied for lane detection. Explain the challenges associated with each processing stage.

Q5. A municipal corporation wants to classify waste such as plastic, paper, and metal using a camera installed above a conveyor belt. The captured images may contain overlapping objects, shadows, reflections, noise, and different object orientations.
Prepare a technical presentation demonstrating a complete image-processing pipeline using preprocessing, enhancement, segmentation, feature extraction, and classification. Explain how the proposed pipeline can be applied to handle real-world variations and improve waste classification.

Q1. Image Enhancement for Night-Time CCTV Surveillance

1. Introduction



A shopping mall uses Closed-Circuit Television (CCTV) cameras to monitor people, objects, entrances, corridors, and parking areas. During the daytime, the cameras usually capture clear images because natural light provides sufficient illumination. However, at night, the captured images may become dark, blurry, and noisy due to insufficient lighting and limitations of the camera sensor.

Low-quality images create difficulties for security personnel and automated object-detection systems. People may not be clearly visible, and objects may blend into the background. Noise may also create false patterns that can be mistaken for actual objects.

Digital image processing provides techniques to improve the quality of captured images before object detection. Filtering, noise removal, brightness correction, contrast enhancement, and sharpening can be combined to produce clearer images.

The main objective is to improve the visibility of people and objects in night-time CCTV footage while preserving important details needed for accurate detection.

2. Objectives

The main objectives of the image-enhancement system are:

- To remove unwanted noise from night-time CCTV images.
- To improve brightness in dark regions.
- To increase contrast between objects and the background.
- To enhance edges and fine image details.
- To improve the quality of images supplied to an object-detection algorithm.
- To support more reliable security surveillance.

3. Filtering Techniques

Filtering is an image-processing operation that modifies pixel values using information from neighbouring pixels. It is commonly used to reduce noise and smooth unwanted variations.

3.1 Mean Filter

A mean filter replaces each pixel with the average intensity of its neighbouring pixels.

For a neighbourhoods containing \(N\) pixels,

\[g(x,y)=\frac{1}{N}\sum_{(i,j)\in S}f(i,j) \]

where:

- \(f(i,j)\) represents the original pixel intensity.
- \(S\) represents the selected neighborhood.
- \(N\) represents the number of pixels.
- \(g(x,y)\) represents the filtered output.

Advantages:

- Simple to implement.
- Reduces small intensity variations.
- Useful for mild noise reduction.

Limitations:

- May blur object boundaries.
- Can remove fine details such as facial features.

3.2 Median Filter

A median filter replaces the center pixel with the median value of the surrounding pixels.

3.3 Gaussian Filter

A Gaussian filter applies weighted averaging, giving more importance to nearby pixels.

It reduces high-frequency noise and produces a smoother image. However, excessive smoothing may remove important object boundaries.

For CCTV images, the filter should be selected according to the type and strength of noise present.

4. Noise Removal

Noise refers to unwanted variations in pixel intensity that reduce image quality. It may occur because of poor illumination, sensor limitations, or increased camera gain.

Common types include:

- Gaussian noise: Random intensity variations.
- Salt-and-pepper noise: Isolated bright and dark pixels.
- Low-light sensor noise: Grainy patterns that become visible in dark regions.

Noise removal should generally be performed before strong contrast enhancement because increasing contrast may also make noise more visible.

An appropriate filter can improve the image while preserving important structures such as human silhouettes and object boundaries.

5. Contrast Enhancement

Contrast enhancement increases the visual difference between dark and bright regions.

5.1 Histogram Equalization

Histogram equalization redistributes intensity values to improve the use of the available brightness range.

It can improve the visibility of objects that are difficult to distinguish from dark backgrounds.

5.2 CLAHE

Contrast Limited Adaptive Histogram Equalization (CLAHE) improves contrast in local regions while limiting excessive enhancement.

It is useful when one part of a CCTV image is dark and another part is already bright.

6. Sharpening Techniques

Sharpening improves the visibility of edges and fine details.

Common techniques include:

- Unsharp masking.
- Laplacian sharpening.
- High-pass filtering.

Unsharp masking adds a controlled detail component back to the original image after subtracting a blurred version. Laplacian sharpening emphasizes rapid intensity changes.

Sharpening can help reveal object boundaries, but excessive sharpening may amplify noise and produce artificial edges.

7. Required Image-Processing Sequence

The recommended sequence is:

1. Input CCTV image

Dark and noisy frame

2. Noise removal

Median or Gaussian filtering

3. Brightness correction

Gamma correction

4. Contrast enhancement

CLAHE or histogram equalization

5. Sharpening

Unsharp masking

6. Object detection

Detect people and objects

The enhanced image is then supplied to an object-detection model. The model identifies objects, predicts their classes, and determines their locations using bounding boxes.

8. Challenges and Solutions

- Very low illumination: Apply brightness correction and suitable low-light enhancement.
- Noise amplification: Perform denoising before strong contrast enhancement.
- Motion blur: Use appropriate camera settings and deblurring methods where feasible.
- Loss of details: Avoid excessive smoothing and sharpening.
- False detections: Evaluate the enhanced images using representative CCTV footage.

9. Applications

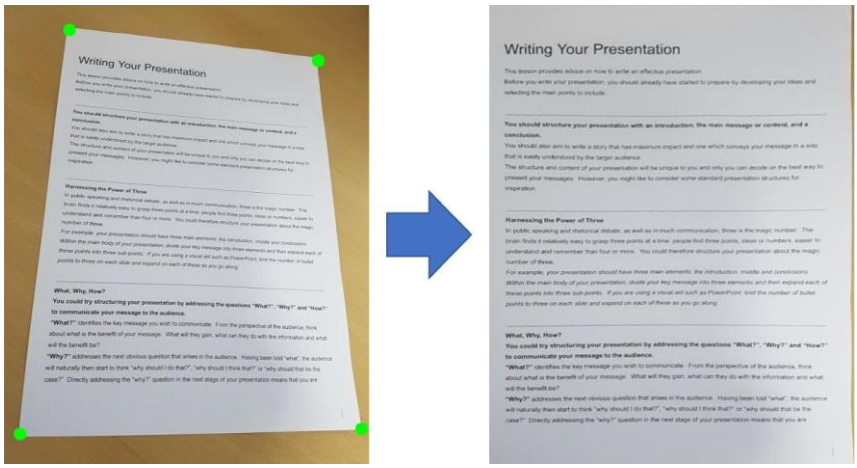
- Shopping mall security.
- Pedestrian monitoring.
- Parking surveillance.
- Intrusion detection.
- Smart building monitoring.

10. Conclusion

Image enhancement improves the quality of night-time CCTV images through filtering, noise removal, brightness correction, contrast enhancement, and sharpening. These techniques prepare images for object detection and may improve the visibility of people and objects. However, preprocessing cannot recover every detail lost through extreme darkness or severe motion blur. Suitable camera placement, adequate lighting, and robust detection algorithms are also essential.

Q2. Image Processing for an Automated Document-Scanning System

1. Introduction



A university handles a large number of documents, including admission forms, examination papers, certificates, application forms, and administrative records. A camera-based document-scanning system can convert these physical documents into digital images for storage, searching, and further processing.

However, captured documents may contain shadows, noise, uneven illumination, tilted pages, and blurred text. These problems reduce the readability of the document and may cause Optical Character Recognition (OCR) software to recognize characters incorrectly.

Digital image processing improves the quality of captured documents by applying filtering, contrast enhancement, thresholding, edge detection, and segmentation. Geometric correction can also be used to straighten tilted pages and correct perspective distortion.

The objective is to produce a clear, properly aligned document image that can be read by humans and processed accurately by OCR software.

2. Objectives

The main objectives are:

- To remove noise from document images.
- To reduce shadows and uneven illumination.
- To improve contrast between text and the background.
- To detect document boundaries.
- To correct tilted pages and perspective distortion.
- To separate text and other meaningful document regions.
- To improve OCR accuracy.

3. Filtering Techniques

Filtering removes unwanted noise while preserving important text details.

3.1 Median Filtering

Median filtering replaces each pixel with the median intensity of its neighboring pixels.

It is particularly useful for removing isolated black or white pixels caused by camera noise.

For example, if a scanned document contains unwanted black dots around printed text, median filtering may remove these dots while preserving many character boundaries.

Advantages:

- Removes certain types of impulse noise.
- Preserves edges better than a simple averaging filter in many cases.
- Useful for cleaning scanned documents.

Limitations:

- Excessive filtering may remove punctuation and small characters.
- Fine text strokes may become distorted.

3.2 Gaussian Filtering

Gaussian filtering smooths the image by assigning higher weights to nearby pixels.

It can reduce mild random noise and make the document background more uniform.

However, strong Gaussian filtering may blur character boundaries and reduce OCR accuracy.

3.3 Bilateral Filtering

Bilateral filtering smooths similar regions while preserving many strong edges.

It may be useful when the paper background contains noise but the printed text must remain sharp.

4. Contrast Enhancement

Contrast enhancement improves the difference between printed text and the paper background.

4.1 Histogram Equalization

Histogram equalization redistributes grayscale intensities to improve overall contrast.

It can make faint text more visible when the original image uses only a limited range of intensities.

However, global enhancement may produce inconsistent results when the document contains both bright and dark regions.

4.2 CLAHE

Contrast Limited Adaptive Histogram Equalization improves contrast locally.

It is particularly useful when:

- One side of the document is darker than the other.
- A shadow covers part of the page.
- Some text regions are faint.
- Lighting varies across the document.

4.3 Illumination Correction

Illumination correction compensates for gradual changes in background brightness.

For example, if the upper portion of a page is bright and the lower portion is gray because of a shadow, background normalization can help make the paper appear more uniform.

This is especially important before thresholding.

5. Thresholding

Thresholding converts a grayscale image into a binary image, commonly representing text in black and the background in white.

6. Edge Detection

Edge detection identifies boundaries by locating sudden changes in pixel intensity.

6.1 Sobel Operator

The Sobel operator calculates intensity gradients in horizontal and vertical directions.

It can identify the boundaries of a document and the edges of printed characters.

6.2 Canny Edge Detection

Canny edge detection involves several operations:

1. Noise reduction.
2. Gradient calculation.
3. Non-maximum suppression.
4. Double thresholding.
5. Edge linking.

Canny edge detection can help identify the outer boundaries of a document.

However, shadows may create false edges, so the detected page boundaries should be checked for consistency.

7. Segmentation and Perspective Correction

Segmentation divides the image into meaningful regions.

In a document-scanning system, segmentation can separate:

- The document from the surrounding background.
- Text from the paper.
- Tables from paragraphs.
- Signatures and stamps from other content.
- Different columns and sections of a form.

After detecting the document boundaries, geometric correction can be applied.

Perspective correction transforms a tilted or angled document into a rectangular view. The four corners of the page are estimated and mapped to the corners of a rectangular output image.

Deskewing can also correct a small rotation so that the text lines become horizontal.

These operations improve readability and help OCR software process the document in the correct orientation.

8. Complete Processing Pipeline

Image acquisition → Grayscale conversion where appropriate → Filtering → Illumination correction → Contrast enhancement → Edge detection → Page-boundary detection → Perspective correction → Thresholding → Segmentation → OCR.

Each operation has a specific role. The pipeline may be adjusted depending on the document type and the quality of the captured image.

9. Challenges and Solutions

- Shadows: Use illumination correction and adaptive thresholding.
- Uneven lighting: Apply local contrast enhancement.
- Blurred text: Use mild sharpening or suitable deblurring methods.
- Tilted pages: Apply deskewing and perspective correction.
- Complex layouts: Use layout-aware segmentation.
- Small characters: Avoid excessive smoothing and use sufficient image resolution.

10. Applications

- University admission systems.
- Examination-paper digitization.
- Certificate scanning.
- Digital libraries.
- Automated form processing.
- Student-record management.

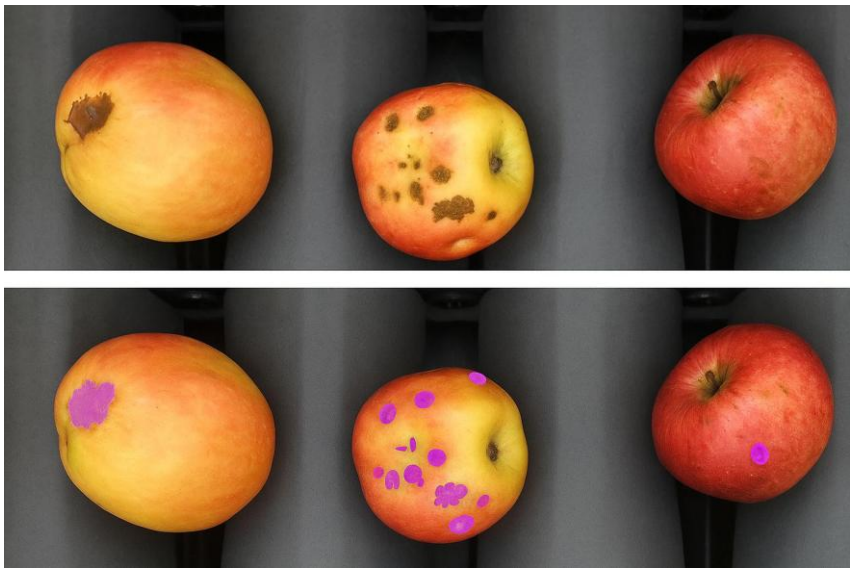
11. Conclusion

An automated document-scanning system combines filtering, contrast enhancement, thresholding, edge detection, segmentation, and geometric correction to improve document quality. Global thresholding is suitable for evenly illuminated pages, whereas adaptive thresholding is more appropriate for uneven lighting. Edge detection helps locate page boundaries, and segmentation separates meaningful regions for further analysis.

The processed image can then be supplied to OCR software to convert printed text into machine-readable form. Appropriate preprocessing improves readability and recognition, although severe blur, missing text, and very low-resolution images may still produce errors.

Q3. Segmentation Techniques for Damaged Apple Detection

1. Introduction



A fruit-processing company wants to automatically identify damaged apples using images captured on a conveyor belt. Damaged apples may contain bruises, dark spots, cuts, rotting regions, or abnormal surface textures.

Manual inspection can be slow and inconsistent, especially when large quantities of fruit must be examined. Computer vision can automate this process by analyzing images and classifying apples according to their visible condition.

However, the captured images may contain shadows, reflections, different backgrounds, and variations in brightness. These conditions make it difficult to distinguish damaged areas from healthy apple skin.

Image segmentation divides an image into meaningful regions. By separating apples from the background and identifying suspicious surface regions, segmentation provides useful information for defect detection.

This answer compares global thresholding, adaptive thresholding, edge-based segmentation, and region-based segmentation.

2. Objectives

- To separate apples from the conveyor-belt background.
- To identify possible damaged regions.
- To compare different segmentation techniques.
- To reduce the effect of shadows and reflections.
- To extract features from suspicious regions.
- To classify apples as healthy or damaged.
- To improve the consistency of automated fruit inspection.

3. Global Thresholding

Global thresholding is a segmentation technique that classifies pixels using one threshold value for the entire image.

Working Procedure

1. Capture an image of the apples.
2. Convert the image to grayscale if intensity-based segmentation is appropriate.
3. Select a threshold value.
4. Classify pixels according to their intensity.
5. Generate a binary mask.
6. Analyze the segmented regions.

Advantages

- Simple to understand and implement.
- Requires relatively little computation.
- Works well when the background is uniform.
- Suitable when the target and background have clearly different intensities.

Limitations

- Sensitive to changes in illumination.
- Shadows may be mistaken for damaged regions.
- Reflections can create misleading bright regions.
- Different apple colors may produce inconsistent segmentation.

Global thresholding is most suitable for controlled environments with uniform lighting and a contrasting conveyor-belt background.

4. Adaptive Thresholding

Adaptive thresholding calculates a local threshold for each region rather than using one value for the entire image.

Working Procedure

1. Capture the apple image.
2. Apply noise reduction.
3. Divide the image conceptually into local neighborhoods.
4. Calculate a threshold for each neighborhood.
5. Generate a binary image.
6. Examine the resulting regions for possible defects.

Advantages

- Handles moderate local brightness variations.

- Can perform better than global thresholding when shadows affect only part of an image.
- Helps identify regions whose intensities differ from their surroundings.

Limitations

- Performance depends on neighborhood size and threshold parameters.
- Surface texture may be incorrectly identified as damage.
- Noise may produce small false regions.
- Healthy and damaged areas with similar intensities may remain difficult to distinguish.

Adaptive thresholding is useful when illumination varies across the conveyor belt, but it should be combined with other features to improve defect classification.

5. Edge-Based Segmentation

Edge-based segmentation identifies boundaries where intensity or color changes significantly.

Common techniques include the Sobel operator and Canny edge detection.

Working Procedure

1. Convert the image into a suitable grayscale or color representation.
2. Reduce noise using filtering.
3. Calculate image gradients.
4. Detect strong edges.
5. Identify contours around apples.
6. Analyze the contours and internal surface regions for possible defects.

6. Region-Based Segmentation

Region-based segmentation groups neighboring pixels with similar properties, such as intensity, color, or texture.

Two common approaches are region growing and watershed segmentation.

6.1 Region Growing

Region growing begins with selected seed points and expands a region by adding neighboring pixels that satisfy a similarity condition.

For example, pixels with similar color and intensity may be grouped into one region.

Advantages:

- Considers relationships between neighboring pixels.
- Identifies connected areas.
- Can help measure the size and shape of damaged regions.

Limitations:

- Results depend on the choice of seed points.
- Similar colors between healthy skin and defects may cause incorrect grouping.

- Shadows and reflections can affect region growth.

6.2 Watershed Segmentation

Watershed segmentation treats an image as a surface containing peaks and valleys. The algorithm uses image gradients and markers to divide the image into regions.

It can be useful for separating touching apples or dividing complex image regions.

However, it may produce too many small regions if the image contains excessive texture or noise. Appropriate preprocessing and markers can reduce this problem.

7. Comparison of Segmentation Techniques

Global thresholding: Fast and simple, but sensitive to illumination changes.

Adaptive thresholding: Handles local brightness variations, but depends on parameter selection.

Edge-based segmentation: Useful for identifying boundaries, but reflections and texture may create false edges.

Region-based segmentation: Groups connected pixels with similar properties, but can over-segment complex surfaces.

Each technique has strengths and limitations. The most suitable method depends on lighting, apple color, background conditions, and defect characteristics.

8. Recommended Combined Processing Pipeline

Image acquisition → Noise removal → Illumination correction → Apple segmentation → Defect-region extraction → Feature extraction → Classification → Inspection report.

Step 1: Image Acquisition

A camera captures images of apples moving along the conveyor belt.

Step 2: Preprocessing

Filtering reduces noise, while illumination correction reduces the effect of shadows and uneven lighting.

Step 3: Apple Segmentation

Color information, thresholding, contour detection, or instance segmentation is used to separate apples from the background.

Step 4: Defect Detection

Adaptive thresholding, color differences, edge information, or region-based methods identify suspicious surface areas.

Step 5: Feature Extraction

Features such as color, area, shape, and texture are calculated for each candidate region.

Step 6: Classification

A classification model determines whether the apple is healthy or damaged.

Step 7: Reporting

The system records the result and may direct damaged apples to a separate inspection or rejection stage.

9. Challenges and Solutions

- Shadows: Use illumination normalization and adaptive thresholding.
- Reflections: Use diffuse lighting and appropriate camera positioning.
- Different apple colors: Combine color information with texture and shape.
- Small defects: Use sufficient image resolution and avoid excessive smoothing.
- Overlapping apples: Use instance segmentation or improve spacing on the conveyor.
- False detections: Evaluate the system using labeled images of both healthy and damaged apples.

10. Applications

- Automated fruit grading.
- Food-processing quality control.
- Fruit packaging industries.
- Agricultural product inspection.
- Reduction of manual inspection workload.

11. Conclusion

Global thresholding, adaptive thresholding, edge-based segmentation, and region-based segmentation are useful techniques for identifying damaged apples. Global thresholding is suitable for uniform conditions, while adaptive thresholding handles local brightness variations. Edge-based segmentation helps identify apple boundaries, and region-based segmentation groups connected areas for further analysis.

A combination of preprocessing, apple segmentation, defect-feature extraction, and classification provides a more robust solution than relying on one technique alone. With suitable lighting, representative training data, and systematic evaluation, automated vision systems can improve fruit inspection and grading.

Q4. Lane Detection for Autonomous Vehicles

1. Introduction



Autonomous vehicles use computer vision to understand road conditions and identify lane markings. Lane detection helps a vehicle estimate its position within a lane and supports driver-assistance functions such as lane-departure warnings and lane-keeping assistance.

A front-facing camera captures images containing road markings, vehicles, shadows, roadside objects, and other environmental features. However, lane detection becomes difficult when the markings are damaged, faded, or covered by vehicles. Rain, reflections, shadows, and changing illumination can also reduce image quality.

Digital image processing techniques such as grayscale conversion, filtering, contrast enhancement, edge detection, thresholding, and region-of-interest selection can help identify lane markings.

The objective is to design a processing pipeline that extracts useful lane information from road-camera images while reducing false detections.

2. Objectives

- To improve the visibility of lane markings.
- To reduce image noise and unwanted road texture.
- To detect possible lane boundaries.
- To distinguish lane markings from irrelevant objects.
- To focus processing on the road region.
- To estimate lane geometry under different environmental conditions.
- To understand the limitations of each processing stage.

3. Grayscale Conversion

Grayscale conversion transforms a color image into an image containing intensity values instead of separate

red, green, and blue channels.

A common conversion formula is:

$$ Y=0.299R+0.587G+0.114B $$

where:

- R represents the red component.
- G represents the green component.
- B represents the blue component.
- Y represents the grayscale intensity.

Advantages

- Reduces the amount of data required for intensity-based processing.
- Simplifies edge detection.
- Highlights brightness differences between road surfaces and lane markings.
- Reduces computational complexity for certain operations.

Challenges

Grayscale conversion removes color information. Yellow lane markings may become difficult to distinguish from other bright road features after conversion.

Solution: Retain the original color image and use color information when it helps identify yellow or white lane markings.

4. Filtering Techniques

Filtering reduces noise and unwanted texture before edge detection.

4.1 Gaussian Filter

Gaussian filtering smooths small intensity variations and reduces high-frequency noise.

It can help remove minor road-surface texture while preparing the image for edge detection.

However, excessive smoothing may remove thin or faded lane markings.

4.2 Median Filter

Median filtering replaces each pixel with the median of the surrounding pixels.

It is useful for removing certain types of isolated noise but may not be sufficient for rain streaks or motion blur.

4.3 Bilateral Filter

Bilateral filtering reduces noise while preserving many strong edges.

It can be useful when lane boundaries must remain visible after smoothing.

Challenges

- Excessive filtering may remove faded lane markings.
- Rain droplets may produce complex image disturbances.

- Motion blur may reduce edge clarity.
- Road texture can create false edges.

The filter must therefore be selected carefully to reduce noise without destroying lane information.

5. Contrast Enhancement

Contrast enhancement improves the difference between lane markings and the road surface.

5.1 Histogram Equalization

Histogram equalization redistributes image intensities to improve global contrast.

It may help make lane markings more visible when the original image has a limited brightness range.

5.2 CLAHE

Contrast Limited Adaptive Histogram Equalization enhances contrast in local regions.

It is useful when part of the road is illuminated while another part is covered by shadows.

5.3 Gamma Correction

Gamma correction adjusts image brightness using a nonlinear transformation. It can help improve the visibility of dark road regions.

However, excessive enhancement may amplify shadows, road texture, and reflections.

Example: A faded white lane marking on a gray road may become easier to distinguish after local contrast enhancement.

6. Edge Detection

Edge detection identifies boundaries where image intensity changes rapidly.

These boundaries may correspond to lane markings, road edges, vehicles, or other objects.

6.1 Sobel Operator

The Sobel operator calculates intensity gradients in horizontal and vertical directions.

It helps identify boundaries and the direction of intensity changes.

6.2 Canny Edge Detection

Canny edge detection consists of the following steps:

1. Smooth the image to reduce noise.
2. Calculate intensity gradients.
3. Apply non-maximum suppression to refine the edges.
4. Use two thresholds to identify strong and weak edges.
5. Connect relevant edges to create an edge map.

The resulting edge map contains candidate lane boundaries.

Challenges

- Shadows can create false edges.
- Road cracks may resemble lane markings.

- Damaged paint may produce discontinuous edges.
- Vehicles can hide portions of the lane.
- Reflections may produce strong but irrelevant boundaries.

Therefore, edge detection must be combined with thresholding, region selection, and geometric analysis.

7. Thresholding

Thresholding selects pixels according to their intensity, color, or gradient strength.

7.1 Intensity Thresholding

Intensity thresholding selects pixels brighter or darker than a specified threshold.

For example, white lane markings may be extracted by selecting sufficiently bright pixels.

However, headlights and road reflections may also be bright.

8. Region of Interest (ROI)

A Region of Interest (ROI) restricts image processing to a selected portion of the image.

For lane detection, the ROI usually covers the road area in front of the vehicle and excludes irrelevant regions such as the sky and buildings.

Working Procedure

1. Define a polygon around the expected road region.
2. Create a binary mask corresponding to that polygon.
3. Apply the mask to the edge or threshold image.
4. Retain candidate lane pixels inside the ROI.
5. Estimate lane boundaries from the remaining pixels.

Advantages

- Reduces irrelevant detections.
- Decreases the amount of image data processed.
- Focuses the algorithm on the road.
- Improves the efficiency of subsequent lane estimation.

Challenges

A fixed ROI may fail when the vehicle travels along a sharp turn, a steep road, or a road with unusual geometry. Camera movement can also change the position of the road in the image.

Adaptive ROI selection can help address these limitations.

9. Complete Lane Detection Pipeline

The recommended pipeline is:

1. Image acquisition: Capture a road image using the front-facing camera.
2. Grayscale conversion: Convert the image to intensity values while retaining useful color information.
3. Filtering: Reduce noise and road-surface texture.

4. Contrast enhancement: Improve the visibility of faded lane markings.
5. Edge detection: Identify candidate lane boundaries.
6. Thresholding: Select relevant pixels using intensity, gradient, or color.
7. ROI selection: Restrict processing to the road region.
8. Lane fitting: Estimate lane geometry using line or curve models.
9. Temporal tracking: Maintain lane estimates across consecutive video frames.
10. Output: Provide lane information to the vehicle's driver-assistance or navigation system.

10. Challenges and Solutions

- Shadows: Use local contrast enhancement and robust geometric fitting.
- Rain: Apply suitable noise reduction and temporal tracking.
- Damaged markings: Combine visible portions of the markings across multiple frames.
- Vehicles blocking lanes: Use tracking and information from visible lane segments.
- Night-time glare: Control exposure and handle bright reflections.
- Curved roads: Use polynomial fitting or more advanced lane models.

11. Applications

- Autonomous driving.
- Lane-departure warning systems.
- Lane-keeping assistance.
- Driver-assistance systems.
- Intelligent transportation systems.
- Road monitoring.

12. Conclusion

Lane detection is an important application of computer vision in autonomous vehicles. Grayscale conversion simplifies intensity processing, filtering reduces noise, contrast enhancement improves visibility, and edge detection and thresholding extract candidate lane markings. ROI selection removes irrelevant regions, while lane fitting and temporal tracking estimate lane geometry.

Each technique has limitations under difficult weather and lighting conditions. Combining these operations can improve lane detection, but safety-critical systems must also use robust perception, monitoring, and appropriate safety mechanisms. Basic image processing alone cannot guarantee reliable lane detection under every road condition.

Q5. Waste Classification Using Image Processing and Computer Vision

1. Introduction



Waste management is an important challenge for municipalities and recycling industries. Materials such as plastic, paper, and metal must be separated before they can be processed and recycled effectively.

Traditional manual sorting can be slow, labor-intensive, and inconsistent. A camera-based computer-vision system installed above a conveyor belt can automatically analyze waste images and classify individual objects according to their material category.

However, real-world waste images may contain overlapping objects, shadows, reflections, noise, and different object orientations. These variations make automated classification more difficult.

A complete image-processing pipeline combines preprocessing, image enhancement, segmentation, feature extraction, and classification. This pipeline prepares the images, separates waste objects, identifies useful visual properties, and assigns each object to a suitable category.

2. Objectives

- To identify waste objects in conveyor-belt images.
- To reduce image noise and unwanted variations.
- To improve contrast and object visibility.
- To separate waste objects from the background.
- To extract color, shape, and texture features.
- To classify objects as plastic, paper, or metal.
- To handle overlapping objects and changing environmental conditions.
- To improve sorting efficiency and recycling operations.

3. Image Acquisition and Preprocessing

Preprocessing prepares captured images for further analysis.

3.1 Image Acquisition

A camera is installed above the conveyor belt to capture images of moving waste objects.

The camera should provide sufficient resolution to reveal relevant details. Suitable focus, exposure, lighting, and camera positioning help reduce the complexity of subsequent processing.

3.2 Grayscale and Color Representation

Grayscale images simplify intensity-based processing, but color information can be important for distinguishing different waste items.

For example, colored plastic packaging and white paper may be easier to distinguish using color features.

Therefore, the original color image should be retained when color is useful for classification, even if certain processing operations use grayscale.

3.3 Filtering

Filtering reduces unwanted noise.

Common methods include:

- Median filtering for isolated noise pixels.
- Gaussian filtering for mild random noise.
- Bilateral filtering for smoothing while preserving many strong edges.

Filtering should not be excessive because surface texture and small boundaries may provide useful information about the material.

4. Image Enhancement

Image enhancement improves the visibility of waste objects and their boundaries.

4.1 Contrast Enhancement

Histogram equalization can improve overall contrast, while CLAHE enhances contrast in local regions.

These techniques can help when some objects appear darker than others because of uneven illumination.

4.2 Illumination Correction

Illumination correction reduces gradual changes in brightness caused by shadows or nonuniform lighting.

For example, one side of a plastic bottle may appear darker than the other. Illumination normalization may reduce this variation.

4.3 Sharpening

Sharpening increases the visibility of object boundaries and surface details.

However, excessive sharpening can amplify noise and create false edges.

Challenges

- Shadows may make objects appear darker.
- Metal reflections can produce very bright regions.

- Plastic and paper may have similar colors.
- Dirty or crushed objects may have unusual textures.

Controlled lighting and suitable camera exposure can reduce these problems before image enhancement is applied.

5. Segmentation

Segmentation separates waste objects from the conveyor-belt background and, where possible, from other objects.

5.1 Threshold-Based Segmentation

Thresholding separates pixels according to intensity.

Global thresholding uses one threshold for the entire image, while adaptive thresholding calculates local thresholds.

These methods can work well when objects contrast clearly with the conveyor belt.

However, thresholding may fail when objects and the background have similar colors or when objects overlap.

5.2 Edge-Based Segmentation

Edge-based segmentation detects boundaries using methods such as Sobel and Canny edge detection.

It can help identify the contours of waste objects.

However, reflections, surface patterns, and overlapping objects may create incomplete or misleading boundaries.

5.3 Region-Based Segmentation

Region-based segmentation groups neighboring pixels with similar properties.

It can help identify connected regions belonging to the same object.

However, crumpled materials, complex textures, and reflections may cause one object to be divided into several regions.

5.4 Object Detection and Instance Segmentation

For complex conveyor environments, deep-learning methods can be more robust than simple thresholding.

Object detection identifies an object's location using a bounding box and predicts its category.

Instance segmentation predicts a separate pixel mask for each detected object.

Instance segmentation is particularly useful when objects overlap because it attempts to distinguish individual items even when they occupy nearby regions.

These methods require suitable labeled training images and may still struggle when objects are heavily occluded.

6. Feature Extraction

Feature extraction converts image information into useful properties for classification.

6.1 Color Features

Color features include average color values and color histograms.

They may help distinguish certain materials or packaging types.

However, color alone is not a reliable indicator of material because plastic, paper, and metal objects can have similar colors.

6.2 Shape Features

Shape features include:

- Area.
- Perimeter.
- Aspect ratio.
- Circularity.
- Contour characteristics.

For example, a flat sheet of paper and a cylindrical metal can may have different shapes.

However, crushed or folded waste may not retain its original shape.

6.3 Texture Features

Texture features describe surface patterns and variations.

They may help distinguish smooth plastic, fibrous paper, and reflective metal.

However, labels, dirt, damage, and lighting can affect the observed texture.

6.4 Deep-Learning Features

A Convolutional Neural Network (CNN) can learn useful visual features directly from labeled images.

Instead of manually defining every feature, a trained model learns combinations of patterns that help distinguish the material categories.

This can be useful when the dataset contains enough representative examples.

7. Classification Techniques

Classification assigns a material category to each detected waste object.

7.1 Support Vector Machine (SVM)

SVM is a supervised machine-learning method that separates classes using a decision boundary.

It can classify waste using extracted color, shape, and texture features.

7.2 Random Forest

Random Forest combines multiple decision trees to make a prediction.

It can classify waste using numerical features and is useful for traditional machine-learning approaches.

7.3 Convolutional Neural Network

A CNN learns visual features from training images and predicts the category of an object.

It may be more effective when material appearance varies considerably across different images.

7.4 Object-Detection Models

Object-detection models predict object locations and class labels.

For overlapping waste, an instance-segmentation model may be more appropriate because it also produces a separate mask for each object.

The system can classify objects into:

- Plastic.
- Paper.
- Metal.

If the model is uncertain, the item can be sent for manual inspection rather than automatically assigned an unreliable label.

8. Complete Image-Processing Pipeline

The proposed pipeline is:

1. Image acquisition: Capture waste images using a conveyor-belt camera.
2. Preprocessing: Resize images and reduce noise.
3. Image enhancement: Improve contrast and normalize illumination.
4. Segmentation: Separate individual objects from the background.
5. Feature extraction: Extract color, shape, texture, or learned features.
6. Classification: Identify plastic, paper, or metal.
7. Sorting: Direct objects to suitable collection bins.
8. Reporting: Record classification results and system performance.

The pipeline combines traditional image processing with machine learning to address the challenges of real-world waste sorting.

9. Handling Real-World Variations

Overlapping Objects

Use instance segmentation to identify individual objects. Suitable camera positioning and spacing on the conveyor can also reduce overlap.

Shadows

Use controlled lighting and illumination normalization to reduce changes in apparent brightness.

Reflections

Diffuse lighting and suitable exposure settings can reduce excessive reflections, especially on metal surfaces.

Different Object Orientations

Train the model using objects in different orientations. Data augmentation with rotations and other suitable transformations can improve robustness.

Similar-Looking Materials

Combine color, shape, texture, and learned features rather than relying on a single property.

Uncertain Predictions

Use a confidence threshold and flag low-confidence objects for additional inspection.

10. Performance Evaluation

The system should be evaluated using labeled images that were not used during training.

Important metrics include:

- Accuracy: The proportion of predictions that are correct.
- Precision: The proportion of predicted class instances that are correct.
- Recall: The proportion of actual class instances that are correctly identified.
- F1-score: The harmonic mean of precision and recall.
- Confusion matrix: Shows which material categories are confused with one another.
- Processing speed: Measures the number of images or objects processed per second.

The evaluation should include overlapping objects, different orientations, shadows, reflections, and damaged waste to determine how well the system performs under real operating conditions.

11. Applications

- Municipal waste segregation.
- Recycling plants.
- Industrial conveyor-belt sorting.
- Material recovery facilities.
- Smart waste-management systems.
- Automated recycling operations.

12. Conclusion

Computer vision can automate waste classification by combining preprocessing, image enhancement, segmentation, feature extraction, and classification. Filtering and enhancement improve image quality, segmentation separates individual objects, and feature extraction provides information about color, shape, and texture.

Traditional image-processing techniques can work well in controlled environments, while deep-learning object detection and instance segmentation can better handle complex backgrounds and overlapping objects when trained on representative data.

A successful waste-classification system requires suitable camera positioning, controlled lighting, accurate training data, systematic evaluation, and appropriate handling of uncertain predictions. By combining these components, the proposed system can improve sorting efficiency and support more effective recycling and waste management.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Content Accuracy	30	Highly accurate, indepth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Faculty In-charge	Course Coordinator	Program Director
Signature: _____ Date: _____	Signature: _____ Date: _____	Signature: _____ Date: _____